Spatially Varying Degradation Modeling and Pansharpening for Large-Scale Multispectral Images

Anjing Guo, *Member, IEEE*, Renwei Dian, *Member, IEEE*, Nan Wang, and Shutao Li, *Fellow, IEEE*

***Abstract*—Existing pansharpening methods mostly rely on globally uniform degradation modeling, such as applying a single MTF-matched image filter or a set of fixed spectral coefficients to the entire image. However, with increasing spatial resolution and coverage of satellite imagery, regional variations in sensor response and atmospheric conditions become non-negligible. These factors introduce spatially varying degradations that cannot be adequately captured by unified models and continuing to use globally uniform degradation models risks introducing spectral or spatial distortions. In this article, we propose an unsupervised pansharpening method based on spatially varying degradation modeling (SVDM). We first build pixel-wise degradation models for the multispectral image (MSI) and panchromatic image (PAN), where each pixel has its own model parameters. Then, we use a fully connected neural network called SVDMNet to efficiently represent them. The SVDMNet also stores the subtle deformation field between MSI and PAN, enabling precise image registration (IR) of the two images. After the training of SVDMNet, we embed it into the classical unsupervised pansharpening framework and freeze its parameters to guide the fusion process of deep pansharpening networks (PSNets). Finally, we construct spatially varying multiloss weights to balance the spatial and spectral loss functions based on the degradation modeling errors. Extensive experiments on various large-scale multispectral datasets demonstrate that the proposed SVDM method can accurately describe complex degradation processes of MSIs and PANs, showing superiority in the pansharpening task of large-scale MSIs and PANs. The code is available at https://github.com/codegaj/SVDM**

***Index Terms*—Anisotropic Gaussian blur kernel, spatially varying degradation modeling (SVDM), unsupervised pansharpening.**

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Anjing Guo, Nan Wang, and Shutao Li are with the School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China, and also with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Changsha 410082, China (e-mail: anjing_guo@hnu.edu.cn; wangn@hnu.edu.cn; shutao_li@hnu.edu.cn).

Renwei Dian is with the School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China, and also with the Innovation Institute of Industrial Design and Machine Intelligence (Quanzhou-Hunan University), Hunan University, Quanzhou 362006, China (e-mail: drw@hnu.edu.cn).

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I. Introduction

BY CAPTURING both detailed spatial and spectral data, multispectral imaging has become an indispensable technology in remote sensing [1]. Higher spatial resolution and wider coverage are two important directions for multispectral imaging. Nowadays, optical satellites are capable of simultaneously acquiring large-scale panchromatic images (PANs) and multispectral images (MSIs) [2]. For example, a rectified standard-scene PAN of the GaoFen-7 satellite contains over $40\,000 \times 40\,000$ pixels, while the MSI contains over $10\,000 \times 10\,000$ pixels. To address the tradeoff between spatial resolution and spectral resolution and to further obtain a higher resolution MSI (HR-MSI), we fuse the acquired PAN and MSI to integrate their spatial and spectral information, which is called pansharpening.

In recent decades, numerous pansharpening methods have been proposed, broadly categorized into four types: component substitution (CS) methods, multiresolution analysis (MRA) methods, variational optimization (VO) methods, and deep learning (DL) methods. CS methods [3], [4], [5], [6], [7], [8], [9], [10] usually utilize a set of spectral coefficients to generate the intensity component of MSI and then replace the intensity component with PAN to reconstruct HR-MSI. Specifically, the IHS method [3] averages spectral bands of MSI to generate the intensity component; the GSA method [5] employs multivariate regression to derive band-specific coefficients, which are then used to construct the intensity component through weighted summation, while the PCA method [6] performs principal component analysis on MSI and considers its first component as the intensity component. CS methods can maintain good spatial details, but they are often accompanied by severe spectral distortions. The core idea of MRA methods [11], [12], [13], [14], [15] is to extract spatial details from PAN and then inject them into the upsampled MSI, with MTF-GLP [11] as a typical method. The most common way to extract spatial details from PAN is filtering PAN with an MTF-matched image filter and then subtracting the filtered image from PAN to obtain residuals. MRA methods have high spectral fidelity, but they are susceptible to the misregistrations between MSI and PAN, causing spatial distortions. VO methods [16], [17], [18], [19], [20], [21], [22], [23], [24] formulate the pansharpening task as a mathematical inverse problem. They construct objective functions based on the relationships between MSI, PAN, and the target HR-MSI and then introduce various priors to solve the problem, with commonly used priors, including sparsity priors [19], [20] and TV prior [25]. VO methods offer high interpretability,

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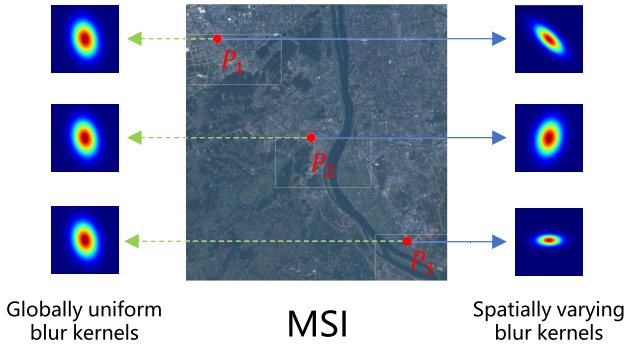


Fig. 1. Comparison of globally uniform blur kernels and spatially varying blur kernels in the large-scale MSI.

but the predefined priors are often insufficient to describe complicated HR-MSI, limiting their performance. Xiao et al. [24] proposed an unsupervised coefficient learning framework, which integrates DL with traditional VO methods and achieved excellent pansharpening performance.

DL methods propose to directly learn the complex mapping between MSI, PAN, and the desired HR-MSI from huge datasets, with fusion architectures being various advanced deep networks, such as CNN-based pansharpening (PNN) [26], PanNet [27], and PanViT [28]. According to different training strategies, DL methods can be further divided into two categories [29]: supervised training methods [26], [27], [28], [30], [31], [32], [33], [34], [35] and unsupervised training methods [36], [37], [38], [39], [40], [41], [42], [43]. Supervised training methods usually use the MTF-matched image filter to blur and downsample the original PAN and MSI to obtain their degraded versions and use MSI as the label to construct training image pairs. Unsupervised methods use the relationships between MSI, PAN, and the target HR-MSI to construct loss functions, which is similar to VO methods. DL methods are currently the most advanced pansharpening branch, with performance significantly better than the aforementioned classical methods.

We usually describe the relationships between MSI, PAN, and the target HR-MSI as degradation models [19], which include the spatial degradation process of MSI and the spectral degradation process of PAN. That is, ideally, MSI can be obtained by spatially blurring and downsampling HR-MSI, while PAN is the weighted summation of all bands of HR-MSI. As shown in Fig. 1, most abovementioned approaches assume that the spatial and spectral degradations of MSI and PAN are globally uniform, i.e., every pixel in the image obeys the same degradation law, and implement pansharpening with a single MTF-matched image filter [44], [45] or a fixed set of coefficients [5]. However, mismatched detector responses between panchromatic and multispectral sensors, platform instabilities, optical distortions, atmospheric effects, and illumination variations in imaging all introduce pronounced and position-dependent differences in the spatial or spectral degradations. For example, in pushbroom spectral systems [46], the spatially varying point spread functions (PSFs) manifest as keystone and smile effects, which arise from spatial misalignments among detector arrays and defective detector pixels, and these artifacts are further amplified by the motion of the imaging platform (e.g., the satellite). Such phenomena are especially prevalent in large-scale remote sensing imagery, rendering conventional globally uniform degradation models inadequate.

In single-image super-resolution, some advanced methods based on spatially varying degradation modeling (SVDM) [47], [48], [49] have been successfully applied. Liang et al. [47] proposed a mutual affine network for spatially varying blur kernel estimation, which leads to state-of-the-art blind super-resolution performance. Chen et al. [49] proposed a novel deep network that simultaneously estimates blur and semantic information, delivering superior performance compared to existing methods. Focusing on the pansharpening task, some early methods based on local regions [8], [50] can be regarded as methods based on SVDM. Garzelli et al. [8] proposed a band-dependent spatial-detail (BDSD) injection model that can be applied to MSI and PAN either globally or locally. Their experimental results show that the HR-MSI obtained with local (spatially varying) estimation is superior to that produced by the global approach. In [50], an improved BDSD method was proposed, which applied a nonlocal parameter optimization through the K-means clustering.

Compared to SVDM in single-image super-resolution, the problem is considerably harder in pansharpening. The complexity arises from the need to jointly model the degradations for both MSI and PAN [51], [52], [53], accounting for their relative spatial positions, spatially varying blur kernels (also called PSFs), and spectral response functions (SRFs). Furthermore, the nonuniform nature of these kernels and functions introduces significant additional computational overhead. Finally, while nearly all existing pansharpening methods rely on globally uniform assumptions, a major unresolved challenge lies in effectively integrating spatially varying models into these algorithms to achieve superior pansharpening performance.

In this article, we propose an SVDM and pansharpening method for large-scale MSIs and PANs. The method contains two parts: SVDM and SVDM-based unsupervised pansharpening. In the SVDM part, we establish spatially varying degradation models for all pixels in PAN and MSI and then introduce a fully connected neural network called SVDMNet to implicitly represent these models. The input of SVDMNet is the coordinates of pixels in MSI, and the output is pixel-wise offsets and parameters of degradation models; hence, we can deal with the misalignments between the two images simultaneously. We train the SVDMNet using the internal relationship between MSI and PAN. In the unsupervised pansharpening part, the trained SVDMNet is incorporated into the classical DL-based unsupervised pansharpening framework and its parameters are frozen. Furthermore, considering the inherent estimation errors of SRFs, we propose a spatially varying multiloss weighting strategy (WS) based on the degradation modeling errors to better maintain the spectral fidelity of fused HR-MSIs. Experiments on several large-scale datasets show that the embedded SVDMNet can better guide the training process of deep pansharpening networks (PSNets), significantly improving the fusion performance. The main contributions of this article are as follows.

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1) We establish spatially varying degradation models for all pixels in large-scale MSIs and PANs and implicitly represent them using a fully connected neural network to avoid enormous memory overhead.
2) We incorporate the spatially varying degradation models into the classical DL-based unsupervised pansharpening framework, improving the performance of various advanced PSNets.
3) Unlike conventional approaches that manually assign weights for spatial and spectral loss functions, we propose a spatially varying multiloss WS based on the degradation modeling errors to better maintain the spectral fidelity of fused HR-MSIs.
4) Experimental results on three large-scale multispectral datasets reveal the existence of spatially varying degradation patterns and validate the superiority of SVDM for pansharpening.

The remainder of this article is organized as follows. In Section II, we review the classical degradation models of pansharpening, implicit neural representation, and DL-based pansharpening methods. The proposed SVDM and pansharpening method is introduced in Section III. In Section IV, extensive experimental results on three large-scale datasets and corresponding discussions are presented. The conclusions and future works are given in Section V.

II. Related Works

A. Classical Degradation Models of Pansharpening

The degradation modeling of pansharpenings describes the relationships between the desired HR-MSI $\mathcal{X} \in \mathbb{R}^{H\times W\times L}$ and the observed MSI $\mathcal{Y} \in \mathbb{R}^{h\times w\times L}$ and PAN $\mathcal{Z} \in \mathbb{R}^{H\times W\times 1}$, where W and H are width and height of the HR-MSI ($h \ll H, w \ll W$), L is the spectral band number, and the fusion ratio is s, $s = W/w$. Classical globally uniform degradation model between HR-MSI and the observed MSI indicates that MSI is a spatial degraded version of HR-MSI, which can be described as

$$\mathcal{Y} = D_{spa}(\mathcal{X}) = (\mathcal{X} * \mathbf{C}) \downarrow \tag{1}$$

where $D_{spa}(\cdot)$ represents the spatial degradation process and $\mathbf{C}$ represents the globally uniform blur kernel, i.e., the relative PSF. * denotes the convolution operation. $\downarrow$ is the spatial downsampling operation, and the downsampling ratio is s. Many studies instantiate the PSF $\mathbf{C}$ as the MTF-matched Gaussian image filter of the multispectral camera.

Similarly, PAN is assumed to be a weighted sum of the bands of HR-MSI, and a globally uniform degradation model describing the relationship between HR-MSI and the observed PAN can be written as

$$\mathcal{Z} = D_{spe}(\mathcal{X}) = \mathcal{X} \times_3 \mathbf{R} \tag{2}$$

where $D_{spe}(\cdot)$ represents this spectral degradation process. $\mathbf{R}$ represents the relative SRF with size $L \times 1$ and it maps the HR-MSI $\mathcal{X}$ to PAN $\mathcal{Z}$, and $\times_3$ represents the tensor-matrix multiplication operation in the spectral dimension. CS methods often use multivariate regression or least squares to get the SRF $\mathbf{R}$.

Furthermore, considering that $D_{spa}(\cdot)$ and $D_{spe}(\cdot)$ are independent of each other, we can obtain

$$D_{spe}\left(D_{spa}(\mathcal{X})\right) \approx D_{spa}\left(D_{spe}(\mathcal{X})\right) \tag{3}$$

i.e.,

$$D_{spe}(\mathcal{Y}) \approx D_{spa}(\mathcal{Z}). \tag{4}$$

This equation is widely used for solving for $\mathbf{R}$ and $\mathbf{C}$ because it only requires the observed MSI $\mathcal{Y}$ and PAN $\mathcal{Z}$.

B. Implicit Neural Representation

Recently, implicit neural networks [54], [55] have shown strong representation and modeling abilities. An implicit neural network is a fully connected neural network, and its powerful representation ability stems from positional-encoding or periodic activation functions. Taking the image representation as an example, the input of the network is the spatial coordinates of pixels, and the output of the network is RGB values of pixels. The above positional encoding can be described as follows:

$$\gamma(x) = \left[\sin\left(2^0\pi x\right), \cos\left(2^0\pi x\right), \ldots, \sin\left(2^{N-1}\pi x\right), \cos\left(2^{N-1}\pi x\right)\right] \tag{5}$$

where x is the row or column coordinate of a pixel, $\gamma(\cdot)$ is the positional-encoding operation, and N is the encoding depth.

The advantage of implicit neural representation is that it can efficiently represent various large and complex fields (a field can be described as a collection of properties for each pixel in the image) using a simple fully connected neural network. This is particularly suitable for handling the deformation field between large-scale MSI and PAN, as well as the parameter fields of spatial and spectral degradation models. Moreover, the smoothness of these fields can be easily controlled by the number of hidden layers and neurons, or the depth of positional encoding. In this article, we even discard the positional-encoding operation to weaken the representation ability of the implicit neural network, thereby maintaining the smoothness of various fields.

C. DL-Based Pansharpening Methods

As shown in Fig. 2, DL-based pansharpening methods can be divided into two categories: supervised training methods [26], [27], [28], [30], [31], [32], [33], [34], [35] on the reduced-resolution images following the Wald protocol [56] and unsupervised training methods [24], [29], [37], [38], [39], [40], [41], [42], [43] on the full-resolution images based on spatial and spectral losses. Supervised training methods based on the Wald protocol typically use an MTF-matched image filter or other Gaussian blur kernel to blur and downsample the original PAN and MSI. The downsampled images serve as the input to the deep network, while the original MSI is used as training labels. Finally, the trained deep network is applied to fuse the original PAN and MSI. Supervised methods focus on designing the structure of the deep PSNet, leading to the development of a number of advanced PSNet architectures, such as convolutional neural networks (CNNs) [26], [27], [30] and vision transformers [28]. Masi et al. [26] proposed the first shallow pansharpening CNN (PNN) for fusing PAN

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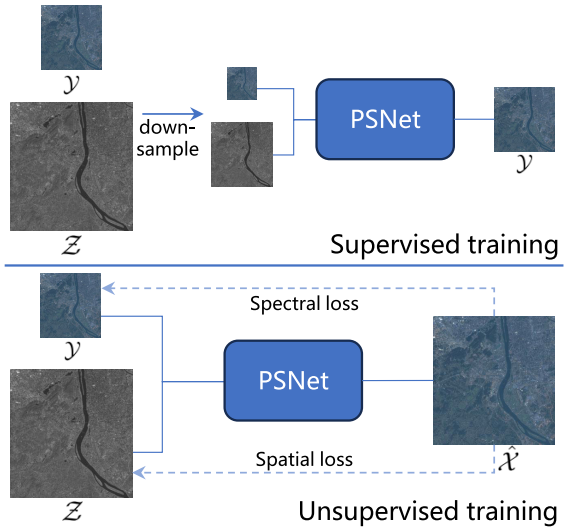


Fig. 2. Comparison of two training strategies of DL methods: supervised training and unsupervised training.

and MSI. This model only consists of three convolutional layers, achieving good pansharpening performance. Yang et al. [27] designed the PanNet architecture by incorporating specific high-frequency domain knowledge of images, and the architecture is trained by inputting residual detail images that have been high-pass filtered. Meng et al. [28] introduced vision transformers into pansharpening to better extract long-range features, which is different from CNNs pursuing a short-distance dependency.

Unsupervised training methods mainly construct spatial and spectral loss functions to constrain the fused HR-MSI, making its spatial and spectral degraded versions close to MSI and PAN, respectively. These methods focus more on the design of unsupervised spatial and spectral loss functions, so many advanced losses have been introduced, such as adversarial loss [37] and perceptual loss [38]. Ma et al. [37] proposed a novel unsupervised pansharpening framework based on generative adversarial network, which could preserve rich spectral information of MSI and spatial information of PAN. Zhou et al. [38] introduced a perceptual loss into the unsupervised pansharpening framework and achieved competitive results over the supervised approaches. Ciotola et al. [41] summarized the aforementioned losses and proposed an improved spectral loss, which significantly enhanced the fusion performance by introducing the ERGAS index. Lin et al. [42] proposed a mutually guided detail restoration method that can adaptively learn spatial and spectral blur kernels directly from the training data. Cao et al. [43] proposed a pansharpening framework with supervised pretraining and unsupervised fine-tuning, which achieved state-of-the-art pansharpening performance. For unsupervised training methods, an additional consideration is how to set the weights for the spatial and spectral losses, i.e., how to balance their respective impacts on the pansharpened HR-MSI.

III. Proposed Method

In this section, we first introduce SVDM, which utilizes a fully connected neural network called SVDMNet to represent various fields. As shown in Fig. 3, the input of SVDMNet is the coordinates of pixels of MSI, and the output of SVDMNet contains pixel-wise offsets and parameters for spatially varying PSFs (SV-PSFs) and spatially varying SRFs (SV-SRFs). As shown in Fig. 4, we then introduce the SVDM-based unsupervised pansharpening, which includes the trained SVDMNet and conventional PSNets, and the parameters of SVDMNet are frozen here.

A. Spatially Varying Degradation Modeling

Compared to classical globally uniform degradation models, spatially varying degradation models refer to each pixel in the image having its own spatial or spectral degradation model. Hence, we establish pixel-wise spatial and spectral degradation models for MSI and PAN. Specifically, the pixel-wise version of spatial degradation model between the desired HR-MSI and the observed MSI can be formulated as

$$\mathbf{y}_{i,j} = D_{spa}\left(\mathcal{X}^{(k)}_{i\cdot s,j\cdot s}\right) = \mathcal{X}^{(k)}_{i\cdot s,j\cdot s} * \mathbf{C}_{i,j} \tag{6}$$

where $D_{spa}(\cdot)$ represents this spatial degradation process, $\mathbf{y}_{i,j}$ is a pixel with size $1 \times L$ in MSI $\mathcal{Y}$, and i and j represent the row and column coordinates of the pixel. $\mathcal{X}^{(k)}_{i\cdot s,j\cdot s}$ is a patch with size $k \times k$ centered at coordinate $(i \cdot s, j \cdot s)$ in HR-MSI $\mathcal{X}$. $\mathbf{C}_{i,j}$ represents the relative PSF of the pixel $\mathbf{y}_{i,j}$, and it has the same size as $\mathcal{X}^{(k)}_{i\cdot s,j\cdot s}$. Hence, the convolution of $\mathcal{X}^{(k)}_{i\cdot s,j\cdot s}$ and $\mathbf{C}_{i,j}$ results in a pixel, that is, $\mathbf{y}_{i,j}$. Furthermore, $\mathbf{C}_{i,j}$ is an anisotropic Gaussian blur kernel [57], [58] with size $k \times k$, which can be formulated as

$$\begin{aligned}\mathbf{C}_{i,j}(m,n) &= \frac{1}{2\pi}\sqrt{|\mathbf{\Lambda}_{i,j}|}\exp\left(-\frac{1}{2}\mathbf{S}^T\mathbf{\Lambda}_{i,j}\mathbf{S}\right)\\ \mathbf{S} &= [m,n]^T, m, n \in -\frac{k-1}{2}, \ldots, \frac{k-1}{2}\end{aligned} \tag{7}$$

where m and n are the coordinates of an element in $\mathbf{C}_{i,j}$. For each $\mathbf{C}_{i,j}$, the parameter matrix $\mathbf{\Lambda}_{i,j} \in \mathbb{R}^{2\times 2}$ is a symmetric positive-definite matrix like $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$, which has three parameters. Inspired by [58], to ensure that the matrix $\mathbf{\Lambda}_{i,j}$ remains symmetric positive definite during the training process, instead of directly optimizing a, b, and c, we represent $\mathbf{\Lambda}_{i,j}$ as follows:

$$\mathbf{\Lambda}_{i,j} = \mathbf{B}_{i,j}\mathbf{B}^T_{i,j} \tag{8}$$

where $\mathbf{B}_{i,j} \in \mathbb{R}^{2\times 2}$ is a lower triangular matrix. Furthermore, we can rewrite $\mathbf{B}_{i,j}$ as

$$\mathbf{B}_{i,j} = \mathbf{M}_{i,j} \odot \mathbf{F} \tag{9}$$

where $\mathbf{F} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $\mathbf{M}_{i,j} \in \mathbb{R}^{2\times 2}$ is what we actually optimize, and $\odot$ represents the Hadamard product. Although $\mathbf{M}_{i,j}$ is a 2×2 matrix, considering the mask matrix $\mathbf{F}$, we argue that $\mathbf{M}_{i,j}$ also has three parameters. It should be noted that although we actually optimize the parameters of $\mathbf{M}_{i,j}$, considering that the shape of $\mathbf{C}_{i,j}$ is directly controlled by the parameters of $\mathbf{\Lambda}_{i,j}$, in subsequent experiments, we only show the results of the three parameters of $\mathbf{\Lambda}_{i,j}$.

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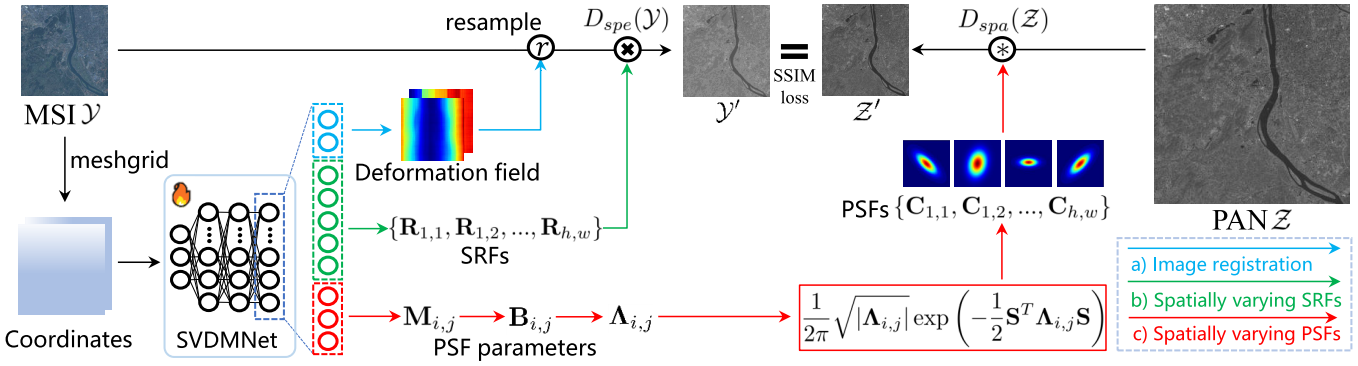


Fig. 3. Flowchart of the proposed SVDMNet and SVDM part. The input of SVDMNet is the coordinates of pixels of MSI, and the output of SVDMNet contains pixel-wise offsets and parameters for SV-PSFs and SV-SRFs.

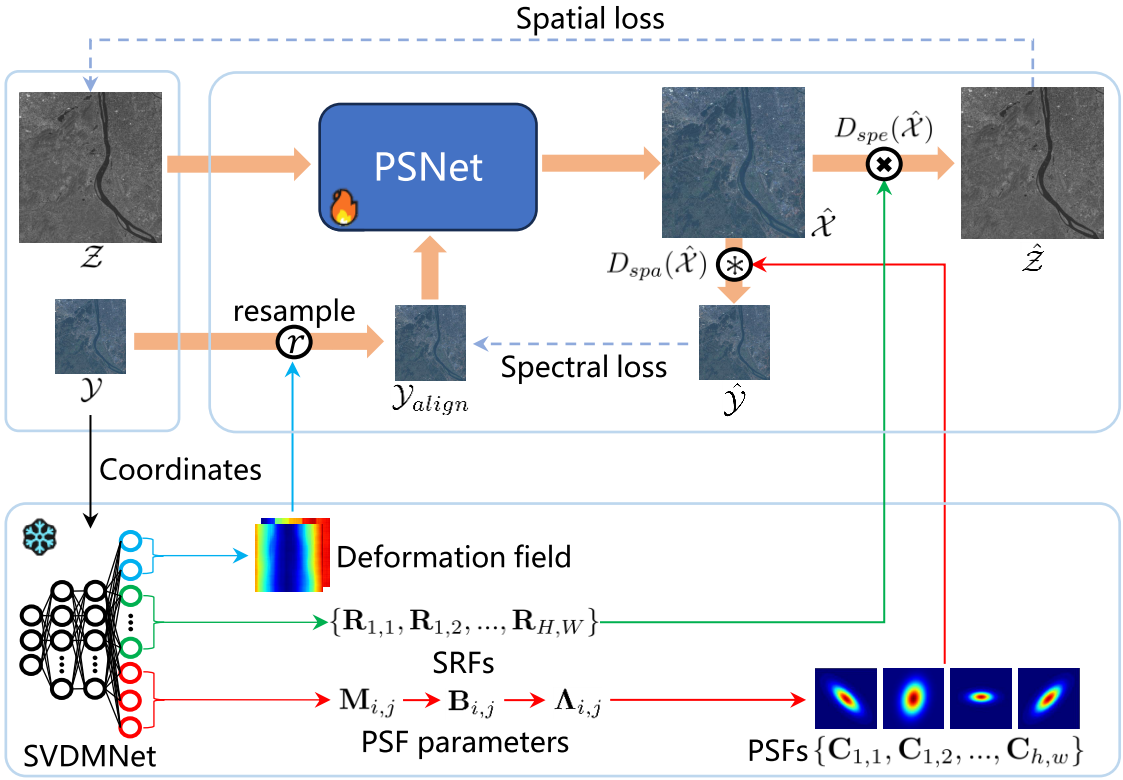


Fig. 4. Flowchart of the proposed SVDM-based unsupervised pansharpening part. The PT SVDMNet is incorporated into the classical unsupervised pansharpening framework, and the parameters of SVDMNet are frozen here.

Furthermore, the spatially varying degradation law of the entire MSI can be represented as the collection of degradation models for all pixels, i.e.,

$$\mathcal{C} = \left\{\mathbf{C}_{1,1}, \mathbf{C}_{1,2}, \ldots, \mathbf{C}_{h,w}\right\}. \tag{10}$$

The pixel-wise version of spectral degradation model describing the relationship between the desired HR-MSI $\mathcal{X}$ and the observed PAN $\mathcal{Z}$ can be written as

$$\mathbf{z}_{i,j} = D_{spe}\left(\mathbf{x}_{i,j}\right) = \begin{pmatrix}\mathbf{x}_{i,j} & 1\end{pmatrix} \times \mathbf{R}_{i,j} \tag{11}$$

where $D_{spe}(\cdot)$ represents this spectral degradation process, $\mathbf{z}_{i,j}$ is a pixel with size 1×1 in PAN $\mathcal{Z}$, and $\mathbf{x}_{i,j}$ is a pixel with size $1 \times L$ in HR-MSI $\mathcal{X}$. To make the modeling results more accurate, we add bias to HR-MSI here, that is, a channel where all elements are 1. $\mathbf{R}_{i,j}$ represents the relative SRF at (i, j) with size $(L + 1) \times 1$.

Similarly, the spatially varying spectral degradation law of the entire PAN can be represented as the collection of degradation models for all pixels, i.e.,

$$\mathcal{R} = \left\{\mathbf{R}_{1,1}, \mathbf{R}_{1,2}, \ldots, \mathbf{R}_{H,W}\right\}. \tag{12}$$

If we represent $\mathcal{R}$ as a field, we can obtain a field tensor of size $H \times W \times (L + 1)$.

Promoting the degradation models (6) and (11) to the entire HR-MSI $\mathcal{X}$, then we can also obtain

$$\mathcal{Y} = D_{spa}\left(\mathcal{X}\right) \tag{13}$$

$$\mathcal{Z} = D_{spe}\left(\mathcal{X}\right). \tag{14}$$

Although the elements in $\mathcal{C}$ and $\mathcal{R}$ are distinct, they actually exhibit certain distributions, e.g., the parameters of these

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degradation models of adjacent pixels should also be close. Hence, if we make

$$\mathcal{Y}' = D_{spe}(\mathcal{Y}) \tag{15}$$

$$\mathcal{Z}' = D_{spa}(\mathcal{Z}) \tag{16}$$

and we will obtain

$$\mathcal{Y}' \approx \mathcal{Z}'. \tag{17}$$

It will be the key to training the SVDMNet.

Unlike the globally uniform degradation models, since the $D_{spe}(\cdot)$ acts on MSI $\mathcal{Y}$ in (15), the field tensor of $\mathcal{R}$ obtained by solving (17) will only have a size of $h \times w \times (L+1)$. Transitioning $\mathcal{R}$ from $h \times w \times (L+1)$ to $H \times W \times (L+1)$ is actually very easy, we can introduce bilinear interpolation, or more effectively, coordinate interpolation (the pixel coordinates of PAN are a fine-grained representation of the pixel coordinates of MSI), which is precisely the advantage of implicit neural representation.

Furthermore, in the process of solving the parameters of $\mathcal{C}$ and $\mathcal{R}$, we also need to consider the subtle offsets between images, which are very common in real scenarios. For example, even after geometric corrections, there are still offsets in several pixels between MSI and PAN. If PAN is regarded as the reference image and MSI is the image to be resampled, it is necessary to provide the offsets of each pixel of MSI in both row and column directions. Hence, we need to construct an additional parameter tensor of deformation field with size of $h \times w \times 2$. When the sizes of PAN and MSI are extremely large-scale, representing and optimizing these parameter tensors and making them smooth will bring a huge computational overhead.

B. SVDMNet and Image Registration

To efficiently represent the deformation field and the parameter fields mentioned above, inspired by implicit neural representation, we introduce a fully connected neural network called SVDMNet to implicitly represent them. As shown in Fig. 3, the structure of SVDMNet is $[2 \to 200 \to 50 \to (2 + L + 1 + 3)]$, the activation function is ReLU for hidden layers, and the output layer is without any activation functions. We input the pixel-wise coordinates of the image into SVDMNet, and the output of SVDMNet consists of three parts: pixel-wise offsets in the row and column directions and parameters of SV-PSF and SV-SRF, which can be represented as

$$\begin{pmatrix} \mathbf{O}_{i,j} \\ \mathbf{R}_{i,j} \\ \mathbf{M}_{i,j} \end{pmatrix} = \text{SVDMNet}(norm(i), norm(j)) \tag{18}$$

where $\mathbf{O}_{i,j}$ represents the offsets of pixel with coordinates (i, j), and $\mathbf{R}_{i,j}$ and $\mathbf{M}_{i,j}$ represent the parameters of degradation models. $norm(\cdot)$ means the normalization operation, and we normalize the coordinates to range $[-1, 1]$.

First, in the image registration (IR) part, the first two neurons of the output layer of SVDMNet provide offsets in the row and column directions, which will be used to resample the MSI. This process can be described as

$$\mathcal{Y}_{align} = Resample(\mathcal{Y}, \mathcal{O}) \tag{19}$$

Listing 1 Implement (21)

```
import torch
import torch.nn.functional as F
def spatially_varying_conv(x, kernels):
    """
    x: [B, C, H, W]
    kernels: [B, H//s, W//s, C, Kh, Kw]
    """

    Kh, Kw = 15, 15 #PSF size
    s = 4 #Fusion ratio
    Ps = Kh // 2 # Padding size

    # Pad and extract patches, start point 1
    x_padded = F.pad(x, [Ps - 1, Ps + 1, Ps - 1, Ps + 1])

    # Use unfold to extract patches
    patches = x_padded.unfold(2, Kh, s).unfold(3, Kw, s)
    patches = patches.permute(0, 2, 3, 1, 4, 5)

    # Element-wise multiplication and sum
    output = torch.sum(patches * kernels, dim=[4, 5])

    return output.permute(0, 3, 1, 2)
```

where $\mathcal{O} = \{\mathbf{O}_{1,1}, \mathbf{O}_{1,2}, \ldots, \mathbf{O}_{h,w}\}$ represents the whole deformation field of MSI and $\mathcal{Y}_{align}$ represents the aligned MSI. The image resampling operation $Resample(\cdot)$ is a differentiable bilinear interpolation, which has been successfully applied in previous works [51], [59].

C. Spectral Degradation

We use the other $L+1$ units of the output layer of SVDMNet to obtain the pixel-wise SRF $\mathbf{R}_{i,j}$ with size $(L+1) \times 1$. As shown in Fig. 3, after the IR process, to obtain $\mathcal{Y}'$, the pixels in the aligned MSI $\mathcal{Y}_{align}$ are then degraded in the spectral dimension, which can be formulated as

$$\mathbf{y}'_{i,j} = D_{spe}\left(\mathbf{y}^a_{i,j}\right) = \begin{pmatrix} \mathbf{y}^a_{i,j} & 1 \end{pmatrix} \times \mathbf{R}_{i,j} \tag{20}$$

where $\mathbf{y}'_{i,j}$ represents the degraded pixel in $\mathcal{Y}'$ and $\mathbf{y}^a_{i,j}$ represents a pixel in the aligned MSI $\mathcal{Y}_{align}$.

D. Spatial Degradation

As mentioned above, the parameter matrix $\mathbf{\Lambda}_{i,j}$ of PSF $\mathbf{C}_{i,j}$ is a symmetric positive-definite matrix; hence, we can use the last three units of the output layer of SVDMNet to generate the pixel-wise spatial degradation model. To obtain $\mathcal{Z}'$, we extract the image patch $\mathcal{Z}^{(k)}_{i\cdot s, j\cdot s}$ in PAN $\mathcal{Z}$ that corresponds to the MSI pixel $\mathbf{y}_{i,j}$ and apply the spatial degradation process to it as

$$\mathbf{z}'_{i,j} = D_{spa}\left(\mathcal{Z}^{(k)}_{i\cdot s, j\cdot s}\right) = \mathcal{Z}^{(k)}_{i\cdot s, j\cdot s} * \mathbf{C}_{i,j} \tag{21}$$

where $\mathbf{z}'_{i,j}$ represents the spatially degraded pixel in $\mathcal{Z}'$ and $\mathcal{Z}^{(k)}_{i\cdot s, j\cdot s}$ is the image patch centered at coordinate $(i \cdot s, j \cdot s)$ in PAN $\mathcal{Z}$.

It should be emphasized that to ensure the efficiency of the training process, we opt for inputting batches of MSI and PAN patches to SVDMNet instead of batches of pixels in code implementation. Compared to the spatially varying spectral degradation process (pixel-wise multiplication), the spatially varying spatial degradation process (patch-wise convolution) is even more complicated. We must crop patches from PAN $\mathcal{Z}$ and convolve it with the corresponding PSFs, and the detailed process is referred to Listing 1. Finally, it is easily observed

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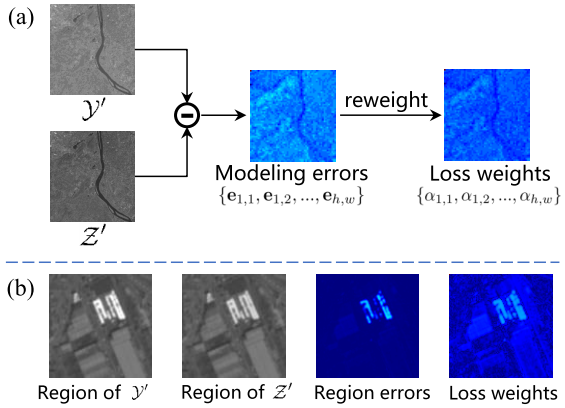


Fig. 5. (a) Flowchart of the proposed loss WS based on the degradation modeling errors. (b) Local region example of the WS. It assigns larger spectral loss weights to regions with significant modeling errors while preventing excessively large weights for outlier pixels.

that in the entire SVDM part, we only need to optimize the parameters of SVDMNet. We introduce the SSIM [60] loss and train SVDMNet with the Adam optimizer [61]. The loss function can be written as

$$\ell_{\text{SVDMNet}} = 1 - \text{SSIM}\left(\mathcal{Y}', \mathcal{Z}'\right). \tag{22}$$

E. SVDM-Based Unsupervised Pansharpening

As shown in Fig. 4, we incorporate the pretrained (PT) SVDMNet into the common unsupervised pansharpening framework and freeze its parameters. For the input MSI $\mathcal{Y}$ and PAN $\mathcal{Z}$, we first query the pixel-wise offsets of MSI from SVDMNet, and then use these offsets to align it with PAN. Then, the aligned MSI and PAN are fed into the deep PSNet to reconstruct the pansharpened $\hat{\mathcal{X}}$, which can be described as

$$\hat{\mathcal{X}} = \text{PSNet}\left(\mathcal{Y}_{align}, \mathcal{Z}\right) \tag{23}$$

where PSNet(·) represents the deep PSNet, which can be instantiated as PNN [26], PanNet [27], and other deep models.

Similarly, we input the pixel-wise coordinates of MSI to query the corresponding PSFs for the spatial degradation process of $\hat{\mathcal{X}}$ and feed the pixel-wise coordinates of PAN to generate the corresponding SRFs for the spectral degradation process of $\hat{\mathcal{X}}$. Although only the coordinates of MSI are used when training SVDMNet, the pixel-wise coordinates of PAN are simply fine-grained representations, and they can be normalized to [−1, 1] and fed directly into SVDMNet. The two degradation processes can be described as

$$\hat{\mathcal{Y}} = D_{spa}\left(\hat{\mathcal{X}}\right) \tag{24}$$

$$\hat{\mathcal{Z}} = D_{spe}\left(\hat{\mathcal{X}}\right) \tag{25}$$

where the efficient implementation of $D_{spa}(\hat{\mathcal{X}})$ is similar to Listing 1. Finally, the obtained $\hat{\mathcal{Y}}$ and $\hat{\mathcal{Z}}$ are utilized to compute the spectral and spatial losses with $\mathcal{Y}_{align}$ and $\mathcal{Z}$, thereby optimizing PSNet.

F. Weighted Loss Functions

Owing to differences in spectral response and coverage between the multispectral and panchromatic sensors [41], [43], we argue that for the spectral degradation model [see (14)] and SRF, the band-weighted summation of the HR-MSI $\mathcal{X}$ can never reproduce $\mathcal{Z}$ perfectly, and irreducible errors remain between $\mathcal{Z}$ and $\hat{\mathcal{Z}}$, especially in overexposed regions such as white roofs or specular reflectors. As shown in Fig. 5, white roofs tend to have larger modeling errors. Furthermore, after accurate IR, the spatial degradation model for $\mathcal{Y}$ [see (13)] is more reliable, that is, a perfectly reconstructed $\hat{\mathcal{X}}$, once spatially degraded, the obtained $\hat{\mathcal{Y}}$ can in principle match $\mathcal{Y}$.

Therefore, we propose that: 1) for the spatial loss between $\mathcal{Z}$ and $\hat{\mathcal{Z}}$, region-based SSIM loss is more appropriate than pixel-based RMSE loss; and 2) for the spectral loss between $\mathcal{Y}_{align}$ and $\hat{\mathcal{Y}}$, it should be assigned a larger weight than the spatial loss, especially in areas where the modeling errors are significant. Specifically, the spatial loss can be expressed as

$$\ell_{spa} = 1 - \text{SSIM}\left(\mathcal{Z}, \hat{\mathcal{Z}}\right). \tag{26}$$

For the weight of spectral loss, as shown in Fig. 5, we propose a spatially varying WS based on the modeling errors as

$$\mathbf{e}_{i,j} = \mathbf{y}'_{i,j} - \mathbf{z}'_{i,j} \tag{27}$$

$$\alpha_{i,j} = ln\left(\beta \cdot \left|\mathbf{e}_{i,j}\right| + 1\right) + 1 \tag{28}$$

where $\mathbf{y}'_{i,j}$ and $\mathbf{z}'_{i,j}$ are the pixels in $\mathcal{Y}'$ and $\mathcal{Z}'$, respectively. Hence, $\mathbf{e}_{i,j}$ with size 1×1 represents the degradation modeling error at pixel coordinate (i, j), and $\mathcal{E} = \{\mathbf{e}_{1,1}, \mathbf{e}_{1,2}, \ldots, \mathbf{e}_{h,w}\}$ represents the modeling errors of the entire image. $\alpha_{i,j}$ denotes the spectral loss weight at (i, j), and we control the range of $\alpha_{i,j}$ by parameter β, expressed concretely as

$$\beta = \frac{2}{\text{MAE}\left(\mathcal{Y}', \mathcal{Z}'\right)}. \tag{29}$$

This ensures that the weights over most image regions remain close to 2 while preventing outlier pixels from excessively large weights. It should be noted that, after the SVDM stage, $\mathcal{Y}'$ and $\mathcal{Z}'$ are fixed; consequently, the value of β is also known.

Finally, we adopt MAE as the basic spectral loss, and the overall loss can be expressed as

$$\ell_{spe} = mean\left(\partial \cdot \left(\left|\mathcal{Y}_{align} - \hat{\mathcal{Y}}\right|\right)\right) \tag{30}$$

$$\ell_{\text{PSNet}} = \ell_{spe} + \ell_{spa} \tag{31}$$

where $\partial = \{\alpha_{1,1}, \alpha_{1,2}, \ldots, \alpha_{h,w}\}$.

IV. Experiments

A. Large-Scale Datasets

In this section, we evaluate the proposed SVDM and SVDM-based pansharpening method on three large-scale multispectral datasets captured by different high-resolution optical satellites, including the GaoFen-7 satellite (GF7), Pléiades-1B satellite (P1B), and IKONOS satellite (IKS). The detailed information of three datasets is shown in Table I. The imaging areas of these datasets are very huge, and all the three PANs contain over 1 billion pixels. The GF7 dataset revealed the Changsha area of Hunan Province, China, containing a PAN with a spatial resolution of 0.65 m and a four-band MSI with a spatial resolution of 2.6 m. The P1B dataset revealed the Kaifeng area of Henan Province, China, containing a PAN with

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TABLE I
DETAILED INFORMATION OF THREE LARGE-SCALE MULTISPECTRAL DATASETS FOR EXPERIMENTAL EVALUATION

Dataset	Images	Image Size	bands	GSD
GF7	MSI	11317 × 10536	4	2.6m
	PAN	45268 × 42144	1	0.65m
P1B	MSI	8000 × 8000	4	2m
	PAN	32000 × 32000	1	0.5m
IKS	MSI	18613 × 3808	4	4m
	PAN	74452 × 15232	1	1m

0.5 m resolution and a four-band MSI with 2 m resolution. The IKS dataset revealed the Fuxin area of Liaoning Province, China, containing a PAN with 1 m resolution and an MSI with 4 m resolution. In the following experiments, we construct training data by nonoverlapping cropping PAN and MSI of each dataset.

B. Compared Methods

In terms of comparison methods, we mainly select three categories. The first category is traditional CS and MRA methods, including EXP [62], BDSD-PC [10], GSA [5], and MTF-GLP [11]. The second category is supervised training-based DL methods, including PNN [26], PanNet [27], FusionNet [30], and WFANet [35], all of which rely on the Wald protocol and MTF-matched image filter. The third category is recent state-of-the-art unsupervised training-based DL methods λ-PNN [41], PAN-MGDR [42], and ZS-PAN [43]. We provide three pansharpening backbone architectures for the proposed SVDM-based unsupervised pansharpening method, namely, the abovementioned PNN, PanNet, and FusionNet, which are denoted as SV-PNN, SV-PanNet, and SV-FusionNet, respectively. The detailed introductions of these methods are as follows.

1) *EXP [62]:* Upsampling MSI using a polynomial interpolator with 23 coefficients.
2) *BDSD-PC [10]:* Fusing MSI and PAN by exploiting the BDSD model with physical constraints.
3) *GSA [5]:* Combining GS transformation and adaptive coefficient regression to achieve pansharpening.
4) *MTF-GLP [11]:* Fusing PAN and MSI using generalized Laplacian pyramid and MTF-tailored image filters.
5) *PNN [26]:* The first pansharpening CNN with three shallow convolutional layers.
6) *PanNet [27]:* A popular pansharpening CNN with deep residual learning and high-pass image filters.
7) *FusionNet [30]:* A classical deep CNN inspired by traditional CS and MRA methods.
8) *WFANet [35]:* A state-of-the-art frequency-based pansharpening transformer.
9) *λ-PNN [41]:* An unsupervised pansharpening method with novel loss functions.
10) *PAN-MGDR [42]:* An unsupervised pansharpening method which can adaptively learn spatial and spectral blur kernels.
11) *ZS-PAN [43]:* A state-of-the-art unsupervised pansharpening framework.

C. Experimental Setting

The proposed SVDM and SVDM-based pansharpening method is implemented with Pytorch [63]. We crop the original MSI and PAN of each dataset into patches and send them into SVDMNet and PSNet. The sizes of MSI and PAN patch are set to 64 × 64 and 256 × 256, respectively, and the cropping stride is set to 256. We reconstruct the full huge HR-MSI using an overlapped image stitching strategy with the fused HR-MSI patches, and the stride of image stitching is set to 216. For the training process of SVDMNet, we train it for 50 epochs with the Adam optimizer [61], and the initial learning rate is set to 0.001. For the training process of PSNet, we also use the Adam optimizer [61] and train it for 50 epochs with an initial learning rate of 0.0005.

For the compared methods, we adopt the same strategy of cropping MSI and PAN into patches of 64 × 64 and 256 × 256, respectively, and then stitching the fused HR-MSI patches together. The cropping strides are the same as in the above experimental settings, while other training details are consistent with the original papers.

D. Evaluation Metrics

The goal of the proposed SVDM and SVDM-based pansharpening method is to explore the degradation models and fusion approach for large-scale MSIs and PANs in real scenarios. Therefore, all experiments in this article are conducted on the full-resolution images and use no-reference image quality evaluation indexes. As the most popular no-reference metrics, D_S and D_λ are widely used in full-resolution experiments. However, they more or less have some limitations. For example, the D_S metric often disagrees with human visual observations [32], [43], [64], while D_λ is very time-consuming when processing massive data due to its complex calculation process. Therefore, in this article, in addition to the classic D_S metric, here we also choose the faster and more robust D_λ^F, D_S^R, and their combination RQNR [64], [65] for full-resolution evaluations. We also use an R-RMSE index to measure the error between the degraded HR-MSI and MSI. The calculation of D_λ^F is as follows:

$$D_\lambda^F = 1 - \mathrm{Q2}^n\left(D_{spa}\left(\hat{\mathcal{X}}\right), \mathcal{Y}\right) \tag{32}$$

where $\mathrm{Q2}^n(\cdot)$ is calculating of the UIQI index [66] between two multiband images and $D_{spa}(\cdot)$ is the abovementioned spatial degradation process. For the comparison methods, we use the MTF-tailored image filter to blur and downsample the HR-MSI to describe this degradation process. For the proposed SVDM, it is consistent with (24). The UIQI index can be expressed as

$$\mathrm{Q2}^n = \frac{\left|\sigma_{I\hat{I}}\right|}{\sigma_I \sigma_{\hat{I}}} \cdot \frac{2\sigma_I \sigma_{\hat{I}}}{\sigma_I^2 + \sigma_{\hat{I}}^2} \cdot \frac{2\left|\bar{I}\right|\left|\bar{\hat{I}}\right|}{\left|\bar{I}\right|^2 + \left|\bar{\hat{I}}\right|^2} \tag{33}$$

where I and $\hat{I}$ are the reference and test images, respectively; σ_I and $\bar{I}$ are the standard deviation and mean of I, respectively; and $\sigma_{I\hat{I}}$ is the covariance between I and $\hat{I}$.

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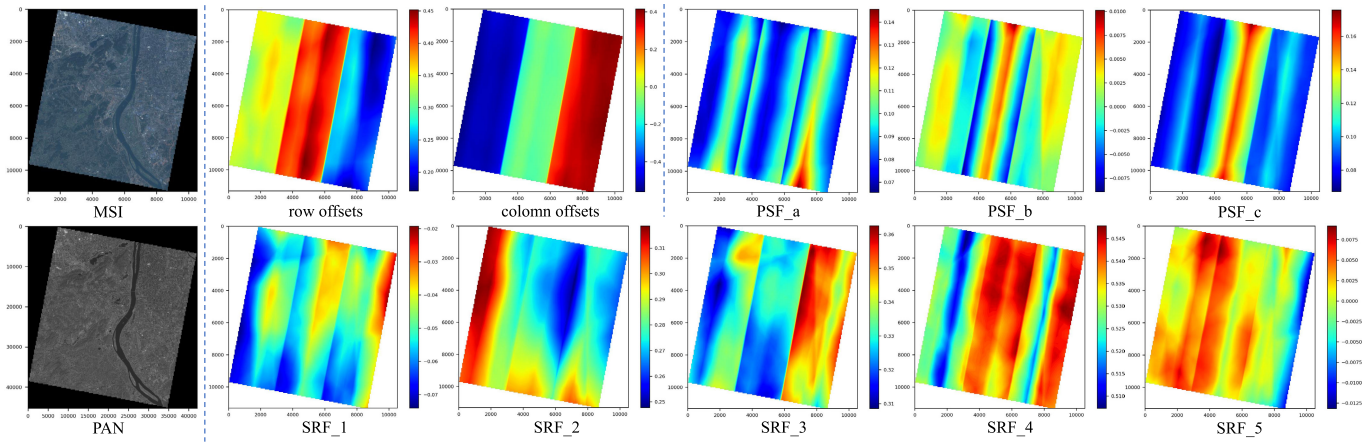


Fig. 6. Visual results of fields of SVDM on the GF7 dataset, including pixel-wise offsets in two directions and parameters of SV-PSF and SV-SRF.

Furthermore, the calculation of D_S^R can be described as

$$D_S^R = \frac{\sigma_\epsilon^2}{\sigma_{\mathcal{Z}}^2} \tag{34}$$

where $\sigma_{\mathcal{Z}}^2$ represents the variance of PAN $\mathcal{Z}$ and σ_ϵ^2 represents the variance of the residual image ϵ. The residual image ϵ can be expressed as

$$\epsilon = \mathcal{Z} - \sum_{i=1}^{L} w_i \hat{\mathcal{X}}_i \tag{35}$$

where the weight w_i of each band of $\hat{\mathcal{X}}$ can be calculated by the optimal minimum mse (mmse) estimation.

The RQNR, which is the combination of D_λ^F and D_S^R, is used to comprehensively evaluate the quality of the pansharpened HR-MSI, i.e.,

$$\text{RQNR} = \left(1 - D_\lambda^F\right)^p \cdot \left(1 - D_S^R\right)^q \tag{36}$$

where the weights p and q are set to 1. Finally, similar to [41], we also propose using the reduced-resolution root-mean-square error (R-RMSE) index to calculate the error between the degraded HR-MSI and the MSI, in order to directly observe the spectral distortions of fused HR-MSIs.

E. Degradation Modeling Results

As mentioned above, the output layer of SVDMNet has ten neurons when L is 4, which represent the pixel-wise offsets, parameters of the spatial degradation kernel PSF, and parameters of the spectral degradation kernel SRF. In this section, we present the visual results of fields composed of these spatially varying parameters. Fig. 6 shows the results on the GF7 dataset, where "row offsets" and "column offsets" represent the offsets of pixels in the MSI, "PSF_{i}" (i = a, b, c) represents the parameters of the symmetric positive-definite matrix $\mathbf{\Lambda}_{i,j}$, and "SRF_{i}" (i = 1, 2, 3, 4, 5) represents the parameters of SRF. The diagonal elements "PSF_a" and "PSF_c" of $\mathbf{\Lambda}_{i,j}$ primarily govern the "spread levels" of the anisotropic Gaussian kernel along the row and column directions, respectively. The larger "PSF_a" and "PSF_c" are, the tighter the PSF distribution; the smaller they are, the looser the PSF distribution. The off-diagonal element "PSF_b" controls the correlation between the row and column directions. If "PSF_b" equals 0, the axes of the PSF are aligned with the coordinate axes; if "PSF_b" is nonzero, the PSF rotates and is no longer axis-parallel. It can be seen that there are obvious strip differences in the row offsets and column offsets of pixels on the GF7 dataset, which may be related to the imaging mode of the GF7 satellite. This also leads to similar variation patterns of PSF and SRF fields. Furthermore, we present typical PSFs and SRFs on the three datasets in Fig. 7. It can be observed that, compared with the GF7 and IKS datasets, the PSFs on the P1B dataset are more consistent, and the PSFs of GF7 and IKS datasets exhibit greater diversity.

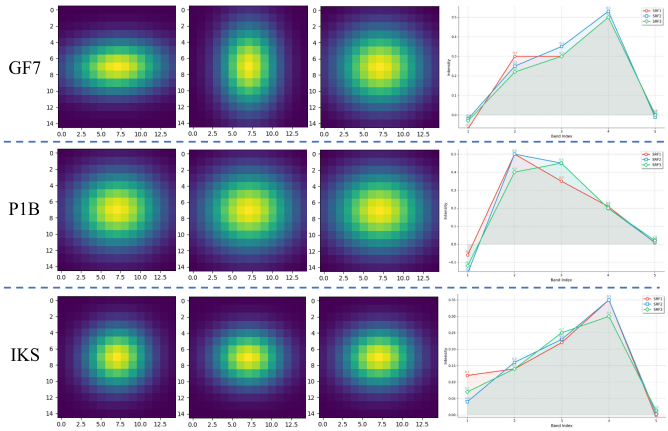


Fig. 7. Visual comparison of modeled representative PSFs and SRFs on three datasets.

F. Quantitative Results

In this section, we present the quantitative results of the proposed method and the compared methods on three large-scale pansharpening datasets. It should be noted that we calculate these metrics patch-wise and then average these values to determine the final scores, and the black border regions of the images do not participate in the metric calculations. The quantitative results on the GF7 dataset are shown in Table II. As can be seen, the deep models based on the proposed SVDM training method achieve better pansharpening results, with lower D_λ^F and D_S^R compared to other methods, which means

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TABLE II
QUANTITATIVE RESULTS (ON D_S, D_λ^F, D_S^R, RQNR, AND R-RMSE) OF THE COMPARED METHODS ON THE GF7 DATASET

Method	D_S	D_λ^F	D_S^R	RQNR	R-RMSE
EXP	0.0645	0.0290	0.1632	0.8100	1.4026
BDSD-PC	0.0950	0.1238	0.0049	0.8721	3.9347
GSA	0.0997	0.0632	0.0048	0.9323	3.0591
MTF-GLP	0.0871	0.0276	0.0401	0.9332	1.3362
PNN	0.0470	0.0256	0.0433	0.9318	1.3542
PanNet	**0.0394**	0.0143	0.0462	0.9400	1.2538
FusionNet	0.0624	0.0314	0.0374	0.9321	1.8543
WFANet	0.0405	0.0235	0.0375	0.9395	1.3986
λ-PNN	0.0499	0.0376	0.0958	0.8692	2.2807
PAN-MGDR	0.1054	0.0634	0.0252	0.9131	2.3246
ZS-PAN	0.0878	0.0146	0.0346	0.9511	1.0723
SV-PNN	0.0586	0.0159	**0.0013**	0.9828	0.6048
SV-PanNet	0.0519	**0.0093**	0.0018	**0.9890**	**0.4908**
SV-FusionNet	0.0541	0.0115	0.0014	0.9871	0.5599

TABLE III
QUANTITATIVE RESULTS (ON D_S, D_λ^F, D_S^R, RQNR, AND R-RMSE) OF THE COMPARED METHODS ON THE P1B DATASET

Method	D_S	D_λ^F	D_S^R	RQNR	R-RMSE
EXP	0.0493	0.0252	0.1606	0.8179	1.2020
BDSD-PC	0.0808	0.0599	0.0107	0.9301	2.3336
GSA	0.1049	0.0340	**0.0041**	0.9621	1.7632
MTF-GLP	0.1040	0.0219	0.0226	0.9561	1.1322
PNN	0.0804	0.0211	0.0105	0.9686	1.0423
PanNet	0.0786	0.0144	0.0142	0.9716	0.9020
FusionNet	0.0815	0.0225	0.0087	0.9691	1.2003
WFANet	0.0525	0.0124	0.0501	0.9382	0.7575
λ-PNN	0.0522	0.0291	0.1318	0.8426	1.6834
PAN-MGDR	0.0897	0.0378	0.0165	0.9464	1.4166
ZS-PAN	0.0692	0.0137	0.0145	0.9721	0.6782
SV-PNN	0.0598	0.0117	0.0043	0.9841	0.5150
SV-PanNet	**0.0484**	**0.0081**	0.0047	**0.9872**	**0.4262**
SV-FusionNet	0.0499	0.0124	0.0042	0.9834	0.5476

TABLE IV
QUANTITATIVE RESULTS (ON D_S, D_λ^F, D_S^R, RQNR, AND R-RMSE) OF THE COMPARED METHODS ON THE IKS DATASET

Method	D_S	D_λ^F	D_S^R	RQNR	R-RMSE
EXP	0.0966	0.0308	0.3343	0.6454	1.2750
BDSD-PC	0.0337	0.0792	0.0332	0.8903	1.7185
GSA	0.0408	0.0385	0.0060	0.9557	1.5088
MTF-GLP	0.0423	0.0197	0.0202	0.9605	0.8346
PNN	0.0265	0.0209	0.0323	0.9475	0.7973
PanNet	0.0294	0.0146	0.0343	0.9516	0.6773
FusionNet	0.0274	0.0242	0.0314	0.9452	0.8398
WFANet	0.0299	0.0122	0.0354	0.9529	0.6179
λ-PNN	0.0282	0.0346	0.0370	0.9297	1.4466
PAN-MGDR	0.0385	0.0593	0.0490	0.8946	1.4093
ZS-PAN	0.0250	0.0135	0.0212	0.9656	0.5923
SV-PNN	0.0332	0.0142	**0.0042**	0.9817	0.5851
SV-PanNet	0.0291	**0.0056**	0.0051	**0.9893**	**0.5132**
SV-FusionNet	**0.0236**	0.0120	0.0045	0.9835	0.5669

TABLE V
QUANTITATIVE SVDM ERROR ANALYSIS OF THREE PROPOSED MODULES: IR, SV-SRF, AND SV-PSF

Dataset	Modules of SVDMNet			SVDM Errors	
	IR	SV-SRF	SV-PSF	RMSE	SSIM
GF7	×	×	×	3.0068	0.9770
	✓	×	×	0.9431	0.9976
	✓	✓	×	0.9408	0.9978
	✓	✓	✓	**0.8416**	**0.9981**
P1B	×	×	×	2.1714	0.9817
	✓	×	×	1.0844	0.9955
	✓	✓	×	0.8577	**0.9957**
	✓	✓	✓	**0.8287**	**0.9957**
IKS	×	×	×	1.0593	0.9922
	✓	×	×	0.7170	0.9957
	✓	✓	×	0.6330	**0.9959**
	✓	✓	✓	**0.6250**	**0.9959**

that our method can significantly reduce spatial and spectral distortions of the pansharpened HR-MSI. Owing to the shallow structure of PNN, the SV-PNN achieves the lowest spatial distortion, but its spectral distortions are relatively severe. In contrast, SV-PanNet, which has a deeper and more complex network structure, obtains better pansharpening results. Among the comparison methods, the unsupervised framework ZS-PAN achieves satisfactory performance. However, due to their use of globally uniform spatial and spectral degradation processes and failure to model the subtle image offsets, their spatial and spectral distortions are still relatively noticeable.

The quantitative results of the P1B and IKS datasets are shown in Tables III and IV, respectively. Similarly, the pansharpening models based on the proposed SVDM method achieve better performance. On the P1B dataset, most of the compared methods demonstrate excellent pansharpening performance. However, thanks to a more flexible SVDM approach, our SVDM method still achieves the lowest D_S, D_λ^F, R-RMSE, and the highest RQNR, which means lower spatial and spectral distortions. Finally, the experimental results on the IKS dataset are consistent with the other two datasets, and the proposed SVDM unsupervised pansharpening method has significant advantages.

G. Visual Comparisons

In this section, we deliver the visual comparisons on three datasets. The visual comparison on the GF7 dataset is shown in Fig. 8. From the magnified image regions, it can be seen that the pansharpening results based on SVDM achieve better image details while maintaining smaller spectral distortion compared to other methods. The BDSD-PC and GSA methods obtain good spatial reconstruction results, but their spectral distortions are relatively severe. The λ-PNN and ZS-PAN methods show some spatial distortions. The visualization results of the P1B and IKS datasets are presented in Figs. 9 and 10, respectively. On these two datasets, almost all the compared methods achieve competitive experimental results. It can be observed that, compared with supervised methods, the unsupervised λ-PNN, ZS-PAN, and SVDM-based pansharpening methods demonstrate better performance in the spectral and spatial reconstruction of vegetation.

H. Effectiveness of SVDM

In this section, we analyze the effectiveness of SVDM. SVDM mainly consists of three modules: accurate IR, SV-SRF, and SV-PSF. We set control groups without the IR

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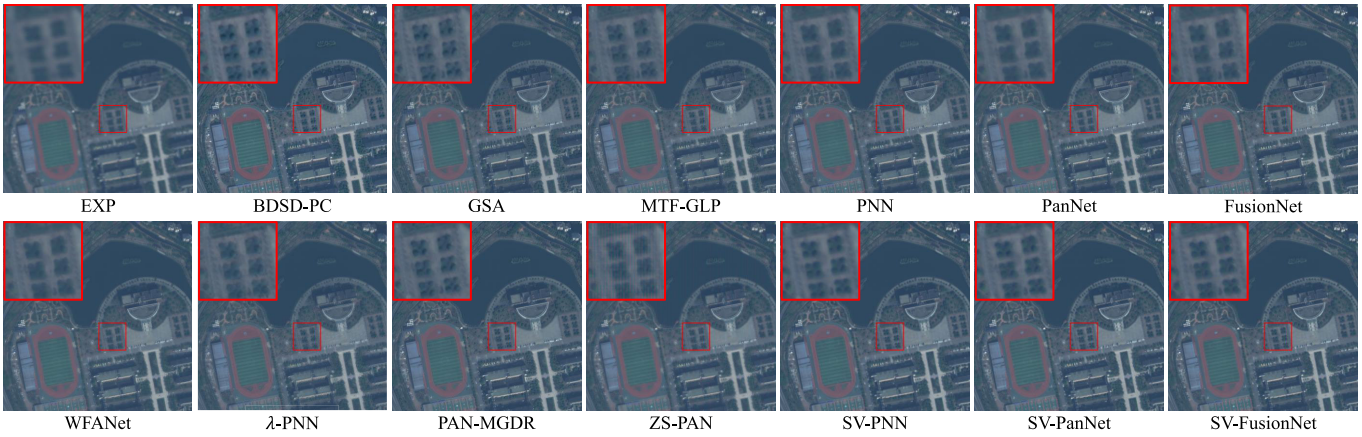


Fig. 8. Visual comparison of full-resolution pansharpening experiments on the GF7 dataset.

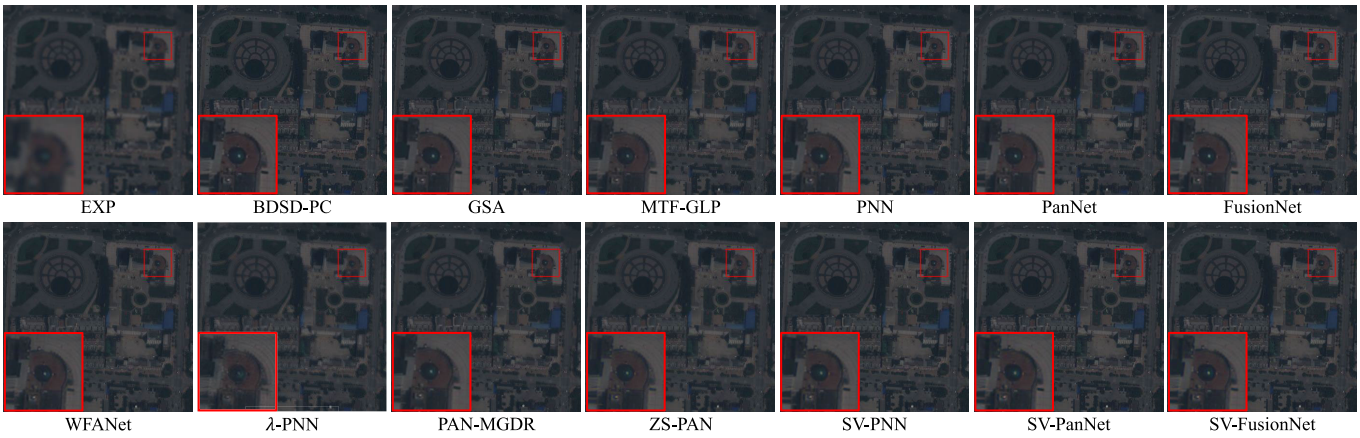


Fig. 9. Visual comparison of full-resolution pansharpening experiments on the P1B dataset.

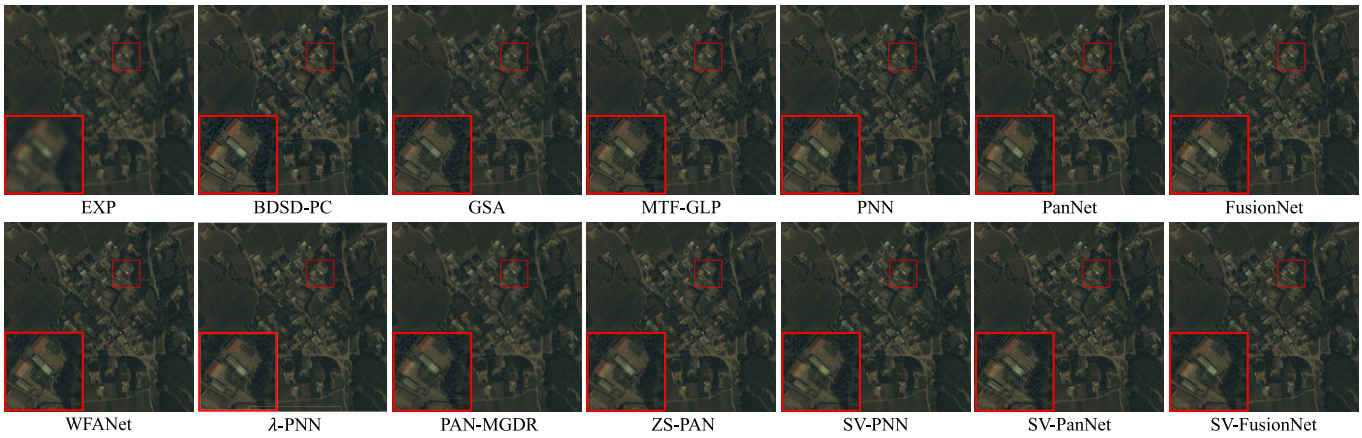


Fig. 10. Visual comparison of full-resolution pansharpening experiments on the IKS dataset.

process and with globally uniform spectral and spatial degradation processes (the groups marked with "×" for IR, SV-SRF, and SV-PSF in Table V), that is, the same estimated SRF and PSF are used for the entire images. Then, we introduce RMSE and SSIM to measure the degradation modeling errors, which means that we calculate the RMSE and SSIM errors between $\mathcal{Y}'$ and $\mathcal{Z}'$, and the experimental results are shown in Table V. It can be seen that the IR module has a significant effect on degradation modeling, even on the previously precisely aligned P1B dataset. The use of SV-SRF and SV-PSF can further improve the degradation modeling performance, which proves the indispensability of the three modules.

I. Effectiveness of PT SVDMNet

As can be seen from Fig. 4, even if we do not freeze SVDMNet in the training process of PSNet, that is, we train

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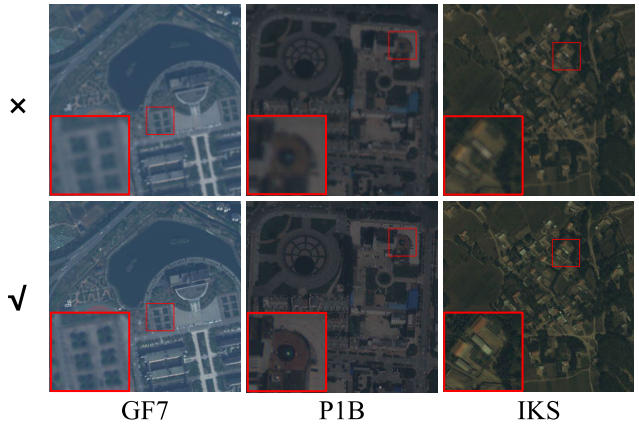


Fig. 11. Visual comparison of the effectiveness of PT SVDMNet with FusionNet on three datasets.

TABLE VI
QUANTITATIVE RESULTS OF SVDM-BASED PANSHARPENING (ON D_λ^F, D_S^R, RQNR, AND R-RMSE) ABOUT PT SVDMNET AND LOSS WS ON THE GF7 DATASET

Backbone	PT	WS	D_λ^F	D_S^R	RQNR	R-RMSE
PNN	×	×	0.0273	0.0082	0.9648	1.4354
	✓	×	0.0175	**0.0008**	0.9817	0.6585
	✓	✓	**0.0159**	0.0013	**0.9828**	**0.6048**
PanNet	×	×	0.0264	0.0132	0.9607	1.8149
	✓	×	0.0137	**0.0015**	0.9847	0.5562
	✓	✓	**0.0093**	0.0018	**0.9890**	**0.4908**
FusionNet	×	×	0.0233	0.0145	0.9626	1.5746
	✓	×	0.0125	**0.0010**	0.9865	0.5887
	✓	✓	**0.0115**	0.0014	**0.9871**	**0.5599**

TABLE VII
QUANTITATIVE RESULTS OF SVDM-BASED PANSHARPENING (ON D_λ^F, D_S^R, RQNR, AND R-RMSE) ABOUT PT SVDMNET AND LOSS WS ON THE P1B DATASET

Backbone	PT	WS	D_λ^F	D_S^R	RQNR	R-RMSE
PNN	×	×	0.0336	0.0121	0.9549	1.1561
	✓	×	0.0128	**0.0038**	0.9835	0.5447
	✓	✓	**0.0117**	0.0043	**0.9841**	**0.5150**
PanNet	×	×	0.0211	0.0108	0.9683	0.9968
	✓	×	0.0105	**0.0044**	0.9852	0.4979
	✓	✓	**0.0081**	0.0047	**0.9872**	**0.4262**
FusionNet	×	×	0.0193	0.0165	0.9646	1.0282
	✓	×	0.0138	**0.0036**	0.9827	0.5729
	✓	✓	**0.0124**	0.0042	**0.9834**	**0.5476**

SVDMNet and PSNet together, the entire process can still be executed. Therefore, we need to verify the necessity of pretraining SVDMNet. We conduct ablation experiments on three datasets using different pansharpening backbones, and the experimental results are shown in Tables VI–VIII, respectively. It can be seen that pretraining SVDMNet and freezing its parameters during the pansharpening process can achieve better fusion performance. The reason is that the degradation modeling process and the unsupervised pansharpening process are highly coupled; if one of them is not fixed and the other process is optimized, there will be a huge solution space, and neither will ultimately achieve the optimal result. Furthermore, we show the visual comparison of the pansharpening results using FusionNet as the backbone in Fig. 11. It can be seen that the PT SVDMNet achieves better results in both spatial details and spectral fidelity.

TABLE VIII
QUANTITATIVE RESULTS OF SVDM-BASED PANSHARPENING (ON D_λ^F, D_S^R, RQNR, AND R-RMSE) ABOUT PT SVDMNET AND LOSS WS ON THE IKS DATASET

Backbone	PT	WS	D_λ^F	D_S^R	RQNR	R-RMSE
PNN	×	×	0.0256	0.0103	0.9644	0.7336
	✓	×	0.0151	**0.0033**	0.9816	0.6213
	✓	✓	**0.0142**	0.0042	**0.9817**	**0.5851**
PanNet	×	×	0.0258	0.0128	0.9617	**0.5017**
	✓	×	0.0085	**0.0043**	0.9872	0.5597
	✓	✓	**0.0056**	0.0051	**0.9893**	0.5132
FusionNet	×	×	0.0177	0.0113	0.9712	0.8213
	✓	×	0.0134	**0.0037**	0.9829	0.6039
	✓	✓	**0.0120**	0.0045	**0.9835**	**0.5669**

J. Effectiveness of Loss WS

Finally, we evaluate the effectiveness of the proposed multiloss WS by conducting ablation experiments on different backbones within the SVDM-based unsupervised pansharpening framework, with quantitative results presented in Tables VI–VIII. It can be observed that the experimental results are consistent with our expectations, i.e., although the introduction of the adaptive multiloss WS slightly increases spatial distortions, we achieve reduced spectral distortions and improved RQNR, effectively ensuring the spectral fidelity of the fused HR-MSI.

V. CONCLUSION

In this article, we mainly analyze the degradation modeling and pansharpening problem of large-size MSIs and PANs, propose a method for spatially varying spatial and spectral degradation modeling, and further apply it to unsupervised pansharpening. Experimental results on large-scale GF7, P1B, and IKS multispectral datasets show that the proposed SVDM method can capture pixel-wise offsets, PSFs, and SRFs, effectively improving the accuracy of degradation modeling. Furthermore, by incorporating the spatially varying spatial and spectral degradation processes into the conventional unsupervised pansharpening framework, the performance of commonly used deep pansharpening models is significantly enhanced. In future works, we will focus on incorporating the spatially varying degradation models into more pansharpening methods and on finding better no-reference evaluation metrics for pansharpened HR-MSIs.

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Anjing Guo (Member, IEEE) received the B.S. and Ph.D. degrees from Hunan University, Changsha, China, in 2016 and 2023, respectively.

He is currently a Post-Doctoral Researcher with the School of Artificial Intelligence and Robotics, Hunan University. His research interests include multi/hyperspectral image processing, image fusion, and deep learning.

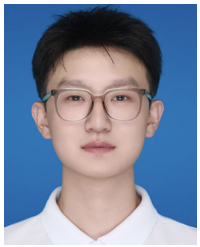
Dr. Guo received the Post-Doctoral Fellowship Program (B Grade) of China Post-Doctoral Researcher Science Foundation in 2024 and was the Champion of the International Algorithm Case Competition hosted by Pazhou Laboratory in 2022.



Renwei Dian (Member, IEEE) received the B.S. degree from Wuhan University of Science and Technology, Wuhan, China, in 2015, and the Ph.D. degree from Hunan University, Changsha, China, in 2020, respectively.

From August 2020 to July 2022, he held a post-doctoral position at the College of Electrical and Information Engineering, Hunan University, where he was awarded the Distinguished Post-Doctoral Fellow. He is currently a Full Professor with the School of Artificial Intelligence and Robotics, Hunan University. His research interests include hyperspectral image super-resolution, image fusion, tensor decomposition, and deep learning.

Dr. Dian was a finalist for the Best Student Paper Award at the International Geoscience and Remote Sensing Symposium (IGARSS) 2018. He was awarded the National Natural Science Award of China, National Natural Science Foundation of China for young Talents, the Wu Wenjun Award for Outstanding Youth in Artifcial Intelligence, the Fellowship of China National Postdoctoral Program for Innovative Talents, and an Excellent Doctoral Dissertation by China Society of Image and Graphics.



Nan Wang received the B.S. and M.S. degrees from Ningbo University, Ningbo, China, in 2020 and 2023, respectively. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence and Robotics, Hunan University, Changsha, China.

His research interests include image fusion, hyperspectral computational imaging, and deep learning.



Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May 2001 to October 2001, He was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He is currently a Full Professor with the School of Artificial Intelligence and Robotics, Hunan University. He has authored or co-authored over 200 refereed articles. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li is a member of the Editorial Board of *Information Fusion* and *Sensing and Imaging*. He gained the 2nd-Grade State Natural Science Award of China in 2019 and two 2nd-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT.

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